SEMTECH UNIFIED SOFTWARE PLATFORM (USP)

Radio Test CLI User Manual

Interactive RF Test Interface for Semtech Radio Modules

LR2021 • LR2022 • SX1261 • SX1262 • SX1268

Version: v0.2.0

Platforms: Linux (Raspberry Pi) · NUCLEO-L476RG · NUCLEO-L073RZ · FPB-RA0E2

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PART 1

Quick Start

Build the CLI, connect your radio, and run your first test.

1.1 Requirements

Semtech radio module: LR2021, LR2022, SX1261, SX1262, or SX1268 connected via SPI/GPIO.

Linux (Raspberry Pi)

ITEM	REQUIREMENT
Host platform	Raspberry Pi with SPI enabled
SPI interface	`/dev/spidev0.0`
GPIO interface	`/dev/gpiochip0`
User groups	User must be in the `spi` and `gpio` groups
Build tools	GCC 12+, CMake 3.25+, Ninja

Baremetal (STM32 / RA0E2)

ITEM	REQUIREMENT
Evaluation board	NUCLEO-L476RG, NUCLEO-L073RZ, or FPB-RA0E2
Build tools	ARM GCC toolchain (`arm-none-eabi-gcc` 12+), CMake 3.25+, Ninja
Serial terminal	ANSI-capable emulator (e.g. Tera Term, PuTTY, minicom) at 115200 baud

1.2 Build

Replace `lr2021` with your chip: `lr2022`, `sx1261`, `sx1262`, or `sx1268`.

Linux (Raspberry Pi)

```
rm -Rf build/
cmake -S examples -B build -DBOARD=LINUX -DRAC_RADIO=lr2021 -G Ninja
cmake --build build --target radio_test_cli
./build/radio_test_cli
```

STM32 (NUCLEO-L476RG or NUCLEO-L073RZ)

```
rm -Rf build/
cmake -S examples -B build -DBOARD=NUCLEO_L476 -DRAC_RADIO=lr2021
-DCMAKE_BUILD_TYPE=MinSizeRel -G Ninja
cmake --build build --target radio_test_cli
```

Replace `NUCLEO_L476` with `NUCLEO_L073` for the L073RZ board.

Renesas FPB-RA0E2

```
rm -Rf build/
env CFLAGS="-DBSP_CFG_HEAP_BYTES=0x400" \
  cmake -S examples -B build -DBOARD=FPB_RA0E2 -DRAC_RADIO=lr2021
-DCMAKE_BUILD_TYPE=MinSizeRel -G Ninja
cmake --build build --target radio_test_cli
```

RA0E2 heap: The `CFLAGS` heap override is required — newlib's `printf` needs heap for its stdio buffer. Without it, the first `printf` call will hard-fault.

GCC 14: Prepend `env CFLAGS="-Wno-incompatible-pointer-types" cmake -UCMAKE_C_FLAGS` to the cmake line to suppress a warning in vendor code.

1.3 First Test: CW in Three Commands

```
region us
power 22
start cw --freq 915.0
```

The radio is now transmitting an unmodulated carrier at 915 MHz, +22 dBm. Type `stop` or press **Ctrl+C** to halt.

What just happened: `region us` loaded US915 defaults. `power 22` set the PA. `start cw` started transmission with a one-shot frequency override. No modulation setup is needed for CW.

The CLI saves history to `.radio_test_history`. Use **Up/Down arrows** to recall previous commands. Press **Tab** to complete commands and parameter values.

PART 2

Test Procedures

Step-by-step guides for each test scenario — from CW power measurements to FCC pre-compliance and PER link tests. Each section explains purpose, parameter choices, and what to expect. For complete parameter syntax, see Part 3.

2.1 CW / Conducted Power Measurement

Purpose: Measure conducted output power and verify frequency accuracy. Also used to confirm PA configuration before modulated testing.

Minimum setup

```
region us
power 22
start cw --freq 915.0
```

No modulation is required for CW. The radio transmits a pure carrier at the specified frequency and power.

What to adjust

PARAMETER	HOW TO SET IT	WHY
`freq`	Band edges and center (e.g. 902.3, 915.0, 927.5 MHz)	Verify power flatness across the band
`power`	Step through the power range	Confirm power accuracy at each level
`pa pa_ramp_us`	`pa pa_ramp_us 16`	Faster ramp; observe switching transients on the analyzer

Inline power sweep (scripted)

```
region us
start cw --freq 915.0 --power 22
delay 10000
stop
start cw --freq 915.0 --power 14
delay 10000
stop
start cw --freq 915.0 --power 0
delay 10000
stop
exit
```

> `--freq` and `--power` on `start` are one-shot overrides. They do not change the stored configuration, so you can sweep without reconfiguring between runs.

2.2 FCC FHSS Test — 64-Channel Hopping

Purpose: Demonstrate compliance with FCC §15.247(a)(1) — frequency hopping spread spectrum across a minimum of 50 channels. The CLI uses the full 64-channel US915 LoRaWAN uplink plan (902.3–914.9 MHz, 200 kHz spacing).

Minimum setup

```
region us
modulation lora
start fhss --dr 0 --count 0
```

This is a complete, valid FHSS test. The radio hops indefinitely through all 64 channels at DR0 (SF10/125 kHz).

Which DR to use

DR	SF	BW	NOTES
0	SF10	125 kHz	Longest packets; easiest for pre-compliance analyzer to trigger on
3	SF7	125 kHz	Shortest packets; fastest hopping throughput
4	SF8	500 kHz	Wider occupied BW per hop

For FCC pre-compliance the test lab will specify which DR to use. DR0 is most commonly requested.

Dwell time

At 125 kHz, DR0 packets are approximately 370 ms. The FCC 400 ms dwell-time limit is not exceeded with back-to-back packets at DR0. The CLI warns if you approach the limit.

Adding inter-hop delay

```
start fhss --dr 0 --count 64 --delay 100
```

Some analyzer setups need settling time between channels. `--delay 100` inserts 100 ms between hops.

Verifying the channel plan

```
show channels
```

Lists all 64 channels and frequencies so you can confirm the plan matches your analyzer's expectations.

2.3 FCC DTS Test — 500 kHz Digital Modulation

Purpose: Demonstrate compliance with FCC §15.247(a)(3) — digital modulation at ≥500 kHz 6 dB bandwidth. DTS mode transmits continuously on a fixed frequency using 500 kHz LoRa bandwidth.

Setup

```
region us
modulation lora
bw 500
sf 7
cr 4/5
pld auto
start dts --freq 915.0
```

Why these parameters:

- `bw 500` — mandatory for DTS; the 6 dB bandwidth must meet the 500 kHz minimum
- `sf 7` — shortest symbols at 500 kHz; maximizes packet rate and minimizes spectral splatter
- `cr 4/5` — minimal overhead
- `pld auto` — fills the packet to maximum size without triggering a dwell-time warning

> **No dwell-time limit applies to 500 kHz DTS operation.** The 400 ms rule applies only to narrower bandwidths.

Varying the test frequency

```
start dts --freq 902.3     # lower band edge
start dts --freq 915.0     # mid-band
start dts --freq 927.5     # upper band edge
```

2.4 FCC Hybrid Test — 8-Channel Sub-Band

Purpose: FCC hybrid mode combines DTS-style 500 kHz LoRa transmission with frequency hopping across an 8-channel sub-band. Used when the product operates as a hybrid FHSS/DTS system under §15.247.

Setup

```
region us
modulation lora
bw 500
sf 7
start hybrid --mask 0 --dr 4 --count 0
```

Sub-band mask reference

MASK	CHANNELS	FREQUENCY RANGE
0	ch 0–7	902.3–903.7 MHz
1	ch 8–15	903.9–905.3 MHz
2	ch 16–23	905.5–906.9 MHz
3	ch 24–31	907.1–908.5 MHz
4	ch 32–39	908.7–910.1 MHz
5	ch 40–47	910.3–911.7 MHz
6	ch 48–55	911.9–913.3 MHz
7	ch 56–63	913.5–914.9 MHz

Why DR4: DR4 is SF8/500 kHz — the natural pairing for 500 kHz hybrid mode. DR3 (SF7/125 kHz) is valid if your product uses narrow-band hopping within the sub-band.

Sweeping all sub-bands (script)

```
region us
modulation lora
bw 500
sf 7
start hybrid --mask 0 --dr 4 --count 100
start hybrid --mask 1 --dr 4 --count 100
start hybrid --mask 2 --dr 4 --count 100
start hybrid --mask 3 --dr 4 --count 100
start hybrid --mask 4 --dr 4 --count 100
start hybrid --mask 5 --dr 4 --count 100
start hybrid --mask 6 --dr 4 --count 100
start hybrid --mask 7 --dr 4 --count 100
exit
```

2.5 PER Link Test

Purpose: Measure packet error rate between two radio units to verify link budget, antenna performance, or range at a given power level.

Setup — Unit A (transmitter)

```
region us
modulation lora
freq 915.0
sf 10
bw 125
power 14
per count 1000
per interval 200
per payload 16
start per-tx
```

Setup — Unit B (receiver)

Configure identically, then:

```
region us
modulation lora
freq 915.0
sf 10
bw 125
start per-rx
```

> **Start the receiver before the transmitter.** Packets transmitted before the receiver is active will not be counted, inflating the PER result.

Reading results

After the test, or at any point during it:

```
per stats
```

```
Last PER test result (rx):
  Received:   997
  CRC errors:   0
  PER:        0.30%
  RSSI avg:  -78 dBm
  SNR avg:   +9.2 dB
```

Sensitivity sweep

To find the sensitivity limit, step power down on the TX side while the receiver runs continuously. Between runs, call `per stats` then `per reset`:

```
per count 500
per interval 100
per payload 16
start per-tx --power 14
start per-tx --power 8
start per-tx --power 2
start per-tx --power -4
exit
```

2.6 FLRC Test (LR2021)

Purpose: Test high-throughput FLRC links or measure FLRC conducted output power. Available on LR2021 only, in `us` and `2g4` regions.

Conducted power / spectral mask

```
region 2g4
modulation flrc
br 1300
cr 3/4
bt bt0.5
pld auto
start modulated
```

Parameter guidance:

- `bt bt0.5` — recommended for spectral mask compliance; reduces out-of-band splatter
- `bt off` — use only if your product disables the filter; widens occupied BW
- `cr none` — maximum throughput, no FEC; most demanding configuration for receiver BER tests

FLRC PER link test

Unit A (TX):

```
region 2g4
modulation flrc
br 2600
cr none
bt off
per count 1000
per interval 10
per payload 255
start per-tx
```

Unit B (RX):

```
region 2g4
modulation flrc
br 2600
cr none
bt off
start per-rx
```

> Both units must use identical `br`, `cr`, and `bt` settings or packets will not be decoded.

PART 3

Command Reference

Complete syntax and valid values for every command and parameter. Use this when you need exact details — Part 2 covers when and why to use them.

3.1 Region and Modulation

`region`

Sets the regulatory region and loads default frequency, power, and channel plan. Must be the first command issued.

```
region <us|2g4>
```

REGION	BAND	FREQUENCY RANGE	DEFAULT FREQ	DEFAULT POWER
`us`	US915	902.0–928.0 MHz	902.3 MHz	+22 dBm
`2g4`	WW-2G4	2400.0–2480.0 MHz	2423.0 MHz	+10 dBm

`modulation`

Sets the modulation type and loads region-appropriate parameter defaults. `modulation help` lists modulations supported by the connected chip.

```
modulation <lora|flrc>
modulation help
```

LoRa is available on all chips. FLRC requires LR2021 in `us` or `2g4`.

`freq`

Sets the transmit/receive center frequency in MHz. Must be within the region's allowed range. Decimal values accepted.

```
freq <MHz>
```

`power`

Sets transmit output power in dBm. Validated against region limits. Updates the PA automatically and **clears all per-parameter PA overrides** (see Section 3.4).

```
power <dBm>
```

`agc`

Controls AGC gain for receive operations. **LR20xx only.**

```
agc <auto|g1|g2|...|g13>
```

VALUE	DESCRIPTION
`auto`	Automatic gain control (default)
`g1` – `g13`	Fixed LNA gain step; `g1` = lowest, `g13` = highest

`boost-lf` and `boost-hf`

Controls the RX boost level for the low-frequency and high-frequency receive paths independently. **LR20xx only.**

```
boost-lf <auto|0-7>
boost-hf <auto|0-7>
```

VALUE	DESCRIPTION
`auto`	Datasheet-recommended default: `0` for LF, `4` for HF
`0` – `7`	Explicit boost level; `7` matches datasheet sensitivity spec condition (G13 rx_boost=7)

Boost only affects gain steps G12 and G13. With `agc auto` (default), the AGC selects G12/G13 on weak signals so boost is active when it matters most for sensitivity. With `agc` fixed to `g1` – `g11`, boost has no effect.

The `status` command marks the currently active path with `[active]` based on the selected frequency (LF below 1.5 GHz, HF at 1.5 GHz and above).

`xosc`

Controls the crystal oscillator load-capacitor trim and stabilization delay. **LR20xx only.**

```
xosc show
xosc default
xosc xta <0-47>
xosc xtb <0-47>
xosc wait <0-255>
```

SUBCOMMAND	DESCRIPTION
`show`	Display current XTA, XTB, and wait values with equivalent pF
`default`	Restore board defaults (xta=0x14, xtb=0x14, wait=150 µs) and apply immediately
`xta <0-47>`	Set XTA pin load-cap trim; C = 11.3 + trim × 0.47 pF
`xtb <0-47>`	Set XTB pin load-cap trim; C = 11.1 + trim × 0.47 pF

SUBCOMMAND	DESCRIPTION
`wait <0-255>`	Set additional XOSC stabilization delay in µs

Each change is applied to the chip immediately. Values accept decimal or `0x`-prefixed hex.

Increase `wait` if CRC errors occur at startup, particularly with narrow-bandwidth FLRC. The default trim values (0x14/0x14) are validated for the FTR5238-A0 crystal (CL=10 pF) with ~2 pF PCB stray. To calibrate for a different crystal, observe the carrier on a spectrum analyser at 2.4 GHz and adjust xta/xtb until the frequency error is within ±1 ppm.

`status`

Displays all current configuration parameters.

```
status
```

3.2 LoRa Parameters

Set after `modulation lora` is active. All values are validated against regional constraints.

`bw` — Bandwidth

```
bw <kHz>
```

REGION	VALID VALUES (KHZ)
US915	125, 250, 500
WW-2G4	812

> Changing bandwidth may invalidate the current SF. The CLI warns when this occurs — use `sf` to adjust before starting a test.

`sf` — Spreading Factor

```
sf <5-12>
```

BW	US915 SF RANGE	NOTES
125 kHz	SF5–SF10	SF10 cap matches LoRaWAN US915 DR0

BW	US915 SF RANGE	NOTES
250 kHz	SF5–SF12	
500 kHz	SF5–SF12	Used for DTS and Hybrid modes

`cr` — Coding Rate

```
cr <4/5|4/6|4/7|4/8>
```

Higher values add FEC at the cost of airtime. Default: `4/5`.

`preamble`

```
preamble <4–65535>
```

Preamble length in symbols. Default: 8 (LoRaWAN standard).

`syncword`

```
syncword <hex>
```

VALUE	NETWORK
`0x34`	LoRaWAN public (US default)
`0x12`	Private

`header`

```
header <explicit|implicit>
```

`explicit` (default) — header carries payload length, CR, and CRC flag. `implicit` — no header; both TX and RX must pre-agree on payload length and CRC.

`crc`

```
crc <on|off>
```

Default: `on`.

`pld` — Payload Size

```
pld <1-255|auto>
```

Sets payload size for `start modulated`. `auto` computes the maximum that fits within the FCC 400 ms dwell-time limit.

`ldro` — Low Data Rate Optimizer

SX126x only (SX1261, SX1262, SX1268).

```
ldro <auto|on|off>
```

VALUE	DESCRIPTION
`auto`	Enabled automatically when symbol duration exceeds 16 ms (default)
`on`	Force LDRO on
`off`	Force LDRO off

3.3 FLRC Parameters

Set after `modulation flrc` is active. **LR2021 only.**

`br` — Bitrate

```
br <kbps>
```

Valid values: 260, 325, 520, 650, 1040, 1300, 2080, 2600. Default: 1300 kbps.

`cr` — Coding Rate

```
cr <1/2|2/3|3/4|none>
```

VALUE	DESCRIPTION
`1/2`	Most redundancy
`2/3`	Moderate
`3/4`	Default
`none`	No FEC — maximum throughput

`bt` — Pulse Shape Filter

```
bt <off|bt0.5|bt1>
```

VALUE	DESCRIPTION
`off`	No filtering
`bt0.5`	Recommended — good spectral containment (default)
`bt1`	Moderate filtering

3.4 PA Configuration

The PA is automatically configured when `power` is set. Use `pa` for fine-grained per-parameter control.

```
pa show
pa reset
pa <param> <value>
```

`pa show` displays current values, valid ranges, and whether each parameter is read-only. `pa reset` clears all per-parameter overrides and restores BSP table values. `pa <param> <value>` overrides a single parameter — other parameters are not affected.

Override semantics: Overrides are per-parameter. Setting one parameter does NOT affect others. Overrides are cleared by:

- `pa reset` — explicit clear
- `power <N>` — recalculates all PA params from the BSP table
- PA path change (LR20xx only) — automatic when switching between LF and HF paths

LR20xx (LR2021, LR2022)

PARAMETER	RANGE	DESCRIPTION	WRITABLE
`half_power`	−39–44	TX power in 0.5 dB steps (LF: −19..44, HF: −39..24)	Yes
`pa_sel`	0–1	PA path: 0=LF (150–960 MHz), 1=HF (1.5–2.5 GHz)	Read-only
`pa_lf_duty_cycle`	0–31	Low-frequency PA duty cycle	Yes
`pa_lf_slices`	0–7	Low-frequency PA number of slices	Yes
`pa_hf_duty_cycle`	16–31	HF PA duty cycle (16=max power, 31=lowest)	Yes

PARAMETER	RANGE	DESCRIPTION	WRITABLE
`pa_ramp_us`	2–304 µs	Ramp time; valid steps: 2, 4, 8, 16, 32, 48, 64, 80, 96, 112, 128, 144, 160, 176, 192, 208, 240, 272, 304	Yes

> `pa_sel` is determined by the operating frequency and cannot be set directly. It is shown in `pa show` for reference.

SX1262 / SX1268

PARAMETER	RANGE	DESCRIPTION	WRITABLE
`power`	−9–22	TX power register value (dBm)	Yes
`pa_duty_cycle`	0–4	PA duty cycle (max 0x04 per datasheet; avoids PA damage)	Yes
`hp_max`	0–7	HP PA output limit; 7 enables +22 dBm	Yes
`pa_ramp_us`	10–3400 µs	Ramp time; valid steps: 10, 20, 40, 80, 200, 800, 1700, 3400	Yes

SX1261

PARAMETER	RANGE	DESCRIPTION	WRITABLE
`power`	−17–14	TX power register value (dBm)	Yes
`pa_duty_cycle`	0–7	PA duty cycle (max 0x04 below 400 MHz, max 0x07 above 400 MHz)	Yes
`pa_ramp_us`	10–3400 µs	Ramp time; valid steps: 10, 20, 40, 80, 200, 800, 1700, 3400	Yes

3.5 Test Modes and Inline Overrides

All modes start with `start <mode>` and stop with `stop` or **Ctrl+C**. Only one mode may be active at a time.

> **MCU plain mode:** During active tick modes (modulated, FHSS, DTS, hybrid, PER), only **Ctrl+C** is recognized — the CLI discards all other input until the mode is stopped. In ANSI mode or on Linux, you can also type `stop`.

`start cw`

Continuous unmodulated carrier.

```
start cw [--freq <MHz>] [--power <dBm>]
```

`start modulated`

Continuous back-to-back packet transmission using the current modulation.

```
start modulated [--freq <MHz>] [--power <dBm>]
```

`start rx`

Continuous receive. Prints RSSI, SNR, and length for each packet.

```
start rx [--freq <MHz>]
```

`start fhss`

US915 64-channel FHSS. US region only.

```
start fhss [--dr <0-6>] [--count <N>] [--delay <ms>] [--power <dBm>]
```

`start dts`

Fixed-frequency 500 kHz LoRa DTS. US region only. BW is forced to 500 kHz.

```
start dts [--freq <MHz>] [--power <dBm>] [--count <N>] [--delay <ms>]
```

`start hybrid`

8-channel sub-band 500 kHz hopping. US region only. `--mask` is required.

```
start hybrid --mask <0-7> [--dr <0-6>] [--count <N>] [--delay <ms>] [--power <dBm>]
```

`start per-tx` / `start per-rx`

PER test transmitter / receiver. Configure with `per` commands before starting (see Section 3.6).

```
start per-tx [--freq <MHz>] [--power <dBm>]
start per-rx [--freq <MHz>]
```

Inline overrides

All `start` commands accept these overrides, which apply to that run only without changing stored configuration:

OVERRIDE	APPLICABLE MODES	DESCRIPTION
`--freq <MHz>`	All	One-shot frequency
`--power <dBm>`	All TX modes	One-shot power
`--dr <0-6>`	fhss, hybrid	Data rate
`--count <N>`	fhss, dts, hybrid	Packet count (0 = ∞)
`--delay <ms>`	fhss, dts, hybrid	Inter-packet delay
`--mask <0-7>`	hybrid (required)	Sub-band selection

3.6 PER Configuration

Configure before `start per-tx` or `start per-rx`. `per stats` and `per reset` are available at any time including during a running test.

COMMAND	DESCRIPTION
`per count <N>`	Packets to send; 0 = infinite (default: 100)
`per interval <ms>`	Inter-packet gap on TX side; must be > 0 (default: 500 ms)
`per payload <6-255|auto>`	Payload bytes; `auto` = max for FCC dwell time (default: 16)
`per stats`	Display last result
`per reset`	Clear last result

3.7 Utility Commands and Scripting

`show channels`

Prints the full channel plan for the current region.

```
show channels
```

`delay <ms>`

Pauses for the given duration. If a test mode is active, it continues running — useful for scripted observation windows. **Linux only.**

`term`

Switches the terminal input mode. **Baremetal (MCU) only** — Linux always uses ANSI mode.

```
term # show current mode
term ansi # enable ANSI terminal (arrow keys, history, tab completion)
term plain # disable ANSI terminal (basic echo mode)
```

MCU builds start in `plain` mode, which uses simple character echo with no escape-sequence processing. This works with any serial terminal. Switching to `ansi` enables the full linenoise line editor with arrow-key navigation, command history, and tab completion, but requires an ANSI-capable terminal emulator (e.g. Tera Term, PuTTY, minicom).

`version`

Prints the CLI version string.

`help [command]`

Full command reference, or detailed help for a specific command.

```
help
help start
help freq
```

`exit` / `quit`

Stops any active test and exits the CLI. **Linux only** — on MCU, reset the board to restart.

Scripting / pipe mode

Linux only. When stdin is not a terminal, the CLI reads commands from stdin and exits at EOF. The scripted examples in Part 2 that use `delay` and `exit` rely on this mode.

```
./build/radio_test_cli <<'EOF'
region us
modulation lora
freq 915.0
power 22
start cw
delay 30000
stop
exit
EOF
```

Exit code: The CLI exits `0` on clean exit. Failed commands print `ERROR:` to stdout but do not abort the script. The overall exit code does not reflect individual command failures.

PART 4

Reference Tables

Region parameters, chip feature matrix, and the complete command quick-reference.

4.1 Region Reference

US915

PARAMETER	VALUE
Frequency range	902.0–928.0 MHz
Default frequency	902.3 MHz (ch0)
TX power range	−10 to +22 dBm
Default TX power	+22 dBm
LoRa BW options	125, 250, 500 kHz
LoRa SF @ 125 kHz	SF5–SF10
LoRa SF @ 250/500 kHz	SF5–SF12
Default SF/BW	SF10 / 125 kHz (DR0)
Default sync word	0x34 (LoRaWAN public)
FLRC	LR2021 only
Regulatory modes	FHSS (64-ch), DTS (500 kHz), Hybrid (8-ch)

US915 Data Rate Reference

DR	SF	BW
0	SF10	125 kHz
1	SF9	125 kHz
2	SF8	125 kHz
3	SF7	125 kHz
4	SF8	500 kHz
5	SF7	500 kHz
6	—	reserved

WW-2G4 (LR2021 only)

PARAMETER	VALUE
Frequency range	2400.0–2480.0 MHz

PARAMETER	VALUE
Default frequency	2423.0 MHz
TX power range	−17 to +12 dBm
Default TX power	+10 dBm
LoRa BW	812 kHz
LoRa SF range	SF5–SF12
FLRC bitrates	260, 325, 520, 650, 1040, 1300, 2080, 2600 kbps
FLRC default	1300 kbps, CR 3/4, BT 0.5
Default sync word	0x21 (WW-2G4 public)
Regulatory modes	None

4.2 Chip Compatibility

FEATURE	LR2021	LR2022	SX1261	SX1262	SX1268
LoRa	✓	✓	✓	✓	✓
FLRC	✓	—	—	—	—
US915 region	✓	✓	✓	✓	✓
WW-2G4 region	✓	—	—	—	—
`agc` command	✓	✓	—	—	—
`boost-lf` / `boost-hf` commands	✓	✓	—	—	—
`xosc` command	✓	✓	—	—	—
`ldro` parameter	—	—	✓	✓	✓
`half_power`, `pa_sel` (r/o)	✓	✓	—	—	—
`pa_lf_duty_cycle`, `pa_lf_slices`, `pa_hf_duty_cycle`	✓	✓	—	—	—
`power` (PA register)	—	—	✓	✓	✓
`pa_duty_cycle`, `hp_max`	—	—	—	✓	✓
`pa_duty_cycle` (LP only)	—	—	✓	—	—
`pa_ramp_us`	✓	✓	✓	✓	✓

FEATURE	LR2021	LR2022	SX1261	SX1262	SX1268
FHSS / DTS / Hybrid	✓	✓	✓	✓	✓

4.3 Quick-Reference Command Table

<table>
<tr><th>COMMAND</th><th>SYNTAX</th><th>DESCRIPTION</th></tr>
<tr><td><code>region</code></td><td><code>region <us|2g4></code></td><td>Set region; load defaults</td></tr>
<tr><td><code>modulation</code></td><td><code>modulation <lora|flrc></code></td><td>Set modulation; load defaults</td></tr>
<tr><td><code>freq</code></td><td><code>freq <MHz></code></td><td>Set center frequency</td></tr>
<tr><td><code>power</code></td><td><code>power <dBm></code></td><td>Set TX power</td></tr>
<tr><td><code>agc</code></td><td><code>agc <auto|g1–g13></code></td><td>AGC gain step (LR20xx only)</td></tr>
<tr><td><code>boost-lf</code></td><td><code>boost-lf <auto|0-7></code></td><td>RX boost level, LF path; auto=0 (LR20xx only)</td></tr>
<tr><td><code>boost-hf</code></td><td><code>boost-hf <auto|0-7></code></td><td>RX boost level, HF path; auto=4 (LR20xx only)</td></tr>
<tr><td><code>xosc</code></td><td><code>xosc <show|default|xta|xtb|wait></code></td><td>Crystal oscillator cap trim and stabilization delay (LR20xx only)</td></tr>
<tr><td><code>bw</code></td><td><code>bw <kHz></code></td><td>LoRa bandwidth</td></tr>
<tr><td><code>sf</code></td><td><code>sf <5-12></code></td><td>LoRa spreading factor</td></tr>
<tr><td><code>cr</code></td><td><code>cr <4/5|4/6|4/7|4/8></code></td><td>LoRa coding rate</td></tr>
<tr><td><code>preamble</code></td><td><code>preamble <symbols></code></td><td>LoRa preamble length</td></tr>
<tr><td><code>syncword</code></td><td><code>syncword <hex></code></td><td>LoRa sync word</td></tr>
<tr><td><code>header</code></td><td><code>header <explicit|implicit></code></td><td>LoRa header mode</td></tr>
<tr><td><code>crc</code></td><td><code>crc <on|off></code></td><td>LoRa CRC</td></tr>
<tr><td><code>pld</code></td><td><code>pld <1-255|auto></code></td><td>Modulated TX payload size</td></tr>
<tr><td><code>ldro</code></td><td><code>ldro <auto|on|off></code></td><td>LoRa LDRO (SX126x only)</td></tr>
<tr><td><code>br</code></td><td><code>br <kbps></code></td><td>FLRC bitrate (LR2021 only)</td></tr>
<tr><td><code>cr</code> (FLRC)</td><td><code>cr <1/2|2/3|3/4|none></code></td><td>FLRC coding rate</td></tr>
<tr><td><code>bt</code></td><td><code>bt <off|bt0.5|bt1></code></td><td>FLRC pulse shape (LR2021 only)</td></tr>
<tr><td><code>pa show</code></td><td><code>pa show</code></td><td>Display PA parameters</td></tr>
<tr><td><code>pa reset</code></td><td><code>pa reset</code></td><td>Clear all PA overrides</td></tr>
<tr><td><code>pa <p> <v></code></td><td><code>pa <param> <value></code></td><td>Override a single PA parameter</td></tr>
</table>

COMMAND	SYNTAX	DESCRIPTION
`status`	`status`	Show all config
`start cw`	`start cw [overrides]`	Unmodulated carrier TX
`start modulated`	`start modulated [overrides]`	Continuous modulated TX
`start rx`	`start rx [overrides]`	Continuous receive
`start fhss`	`start fhss [overrides]`	64-ch FHSS TX (US only)
`start dts`	`start dts [overrides]`	DTS 500 kHz TX (US only)
`start hybrid`	`start hybrid --mask N [overrides]`	Hybrid 8-ch TX (US only)
`start per-tx`	`start per-tx [overrides]`	PER transmitter
`start per-rx`	`start per-rx [overrides]`	PER receiver
`stop`	`stop`	Stop active mode
`per count`	`per count <N>`	PER packet count (0=∞)
`per interval`	`per interval <ms>`	PER inter-packet gap
`per payload`	`per payload <6-255|auto>`	PER payload size
`per stats`	`per stats`	Show last PER result
`per reset`	`per reset`	Clear PER result
`show channels`	`show channels`	Display channel plan
`delay`	`delay <ms>`	Pause (scripting; Linux only)
`term`	`term <ansi|plain>`	Switch terminal mode (MCU only)
`version`	`version`	Show CLI version
`help`	`help [command]`	Show help
`exit` / `quit`	`exit`	Exit CLI (Linux only)

PART 5

Troubleshooting

Common errors and how to resolve them.

Troubleshooting

SPI device not found

```
ERROR: Failed to open SPI device /dev/spidev0.0
```

- Check: `ls /dev/spidev*`
- Enable SPI via `sudo raspi-config` → Interface Options → SPI, then reboot

Permission denied on SPI or GPIO

```
ERROR: Permission denied opening /dev/spidev0.0
```

```
sudo usermod -aG spi,gpio $USER
```

Log out and back in for the group change to take effect.

Build fails on GCC 14

Vendor headers produce incompatible pointer type warnings treated as errors:

```
env CFLAGS="-Wno-incompatible-pointer-types" cmake -UCMAKE_C_FLAGS \
    -S examples -B build -DBOARD=LINUX -DRAC_RADIO=lr2021 -G Ninja
```

Radio not responding / no RX packets

- Run `status` on both units and compare all parameters
- Confirm frequency, BW, SF, CR, syncword, header mode, and CRC all match
- Start `start per-rx` before `start per-tx`

FLRC: "not supported by this chip"

FLRC requires LR2021. Verify you built with `-DRAC_RADIO=lr2021` and the correct module is connected.

AGC command not recognized

`agc` is LR20xx only and is not present in SX126x builds.